📘 AI/ML Internship – Day 78 Notes & Tasks

🟢 Module 9: Conversation Memory & Chat History in RAG – Building Intelligent AI Assistants That Remember Conversations

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📌 Part 1: Recap of Day 77

Yesterday we learned:

- ✅ Metadata
- ✅ Metadata Filtering
- ✅ Source Citation
- ✅ Enterprise Search
- ✅ Trustworthy AI Responses

Today we will learn:

- What is Conversation Memory?
- Why Memory is Needed
- Types of Memory
- Chat History
- Memory in LangChain
- Memory in RAG Systems
- Memory Challenges

📌 Part 2: What is Conversation Memory?

Definition

Conversation Memory is the ability of an AI chatbot to remember previous messages in a conversation and use them to answer future questions correctly.

Simple Definition

Conversation Memory allows the chatbot to **remember what the user said earlier**, just like a human remembers an ongoing conversation.

Example Without Memory

User: My name is Rahul.

Bot: Nice to meet you.

User: What is my name?

Bot: I don't know.

❌ The chatbot forgot the previous message.

Example With Memory

User: My name is Rahul.

Bot: Nice to meet you, Rahul.

User: What is my name?

Bot: Your name is Rahul.

✅ The chatbot remembers the conversation.

📌 Part 3: Why is Memory Important?

Imagine a customer support chatbot.

Conversation:

User: I want to book a flight.

Bot: Which city do you want to travel to?

User: Mumbai.

Without memory:

❌ The chatbot does not know that "Mumbai" is the destination.

With memory:

✅ The chatbot understands that the user is answering the previous question.

📌 Part 4: Conversation Flow

User Message

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Conversation Memory

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Previous Chat History

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Retriever

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▼

Relevant Documents

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▼

LLM

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▼

Final Response

📌 **Part 5: Types of Memory**

1. Buffer Memory

Stores all previous messages.

Example:

Hello

↓

My name is Rahul

↓

I live in Kochi

↓

What is my name?

The chatbot can answer because the full conversation is stored.

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2. Window Memory

Stores only the last **N** conversations.

Example:

Last 5 messages only.

Older messages are removed.

Useful for long conversations.

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3. Summary Memory

Instead of storing every message, the chatbot creates a summary.

Example:

Original Conversation:

100 messages

Summary:

User is asking about internship registration.

This saves memory and reduces token usage.

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4. Entity Memory

Stores important information about specific entities.

Example:

User Name

Rahul

Company

ABC Technologies

Course

AI Internship

The chatbot remembers these important details.

📌 Part 6: Chat History

Every conversation has a **chat history**.

Example:

User:

What is AI?

Assistant:

AI stands for Artificial Intelligence.

User:

Give an example.

Assistant:

ChatGPT is an example of AI.

The chatbot uses this history to answer follow-up questions.

📌 Part 7: Conversation Memory in LangChain

LangChain provides memory classes to store conversations.

Example:

```
from langchain.memory import ConversationBufferMemory


memory = ConversationBufferMemory()
```

This stores the complete conversation.

Window Memory Example

```
from langchain.memory import ConversationBufferWindowMemory


memory = ConversationBufferWindowMemory(k=5)
```

Only the last five conversations are remembered.

📌 Part 8: Memory in RAG

Memory and Retrieval work together.

Flow:

User Question

↓

Conversation Memory

↓

Retriever

↓

Vector Database

↓

Relevant Context

↓

LLM

↓

Answer

The chatbot uses:

- Previous conversation
- Retrieved documents

Both are combined before generating the answer.

📌 Part 9: Why Memory is Useful in RAG?

Suppose the user asks:

Question 1:

Explain Machine Learning.

Question 2:

Give three examples.

Without memory:

The chatbot doesn't know examples of what.

With memory:

The chatbot understands that the user is asking for examples of Machine Learning.

📌 Part 10: Challenges

- Long conversations increase token usage.
- Old information may become irrelevant.
- Memory management becomes difficult.
- Privacy must be protected.

📌 Part 11: Best Practices

✅ Store only useful information.

✅ Remove unnecessary chat history.

✅ Use summary memory for long conversations.

✅ Protect user privacy.

✅ Combine memory with retrieval carefully.

📌 Part 12: Real-World Applications

Customer Support

Remember previous customer issues.

Healthcare

Remember patient history during a session.

Banking

Remember account-related queries.

Education

Remember previous student questions.

Virtual Assistants

Continue natural conversations.

📌 **Part 13: Advantages**

✅ Natural conversations.

✅ Better user experience.

✅ Context-aware responses.

✅ Fewer repeated questions.

✅ More intelligent AI assistants.

📌 **Part 14: Limitations**

❌ Large memory increases API cost.

❌ Irrelevant history may confuse the model.

❌ Sensitive information must be handled securely.

📌 **Part 15: Theory Questions**

1. What is Conversation Memory?

Conversation Memory is the ability of an AI system to remember previous messages and use them in future responses.

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2. Why is Conversation Memory important?

It helps the chatbot understand follow-up questions and maintain context throughout the conversation.

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3. Name four types of memory.

- Buffer Memory
- Window Memory
- Summary Memory
- Entity Memory

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4. What is Buffer Memory?

Buffer Memory stores the complete conversation history.

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5. What is Summary Memory?

Summary Memory stores a summarized version of long conversations to reduce token usage.

📌 **Part 16: Interview Questions**

What is the role of Conversation Memory in RAG?

Conversation Memory allows the chatbot to remember previous interactions while also retrieving relevant information from documents, resulting in more contextual and accurate answers.

Why do we use Window Memory?

Window Memory limits the stored conversation to the most recent messages, reducing token usage while maintaining relevant context.

What is the difference between Memory and Retrieval?

Memory	Retrieval
Stores previous conversation	Retrieves external document information
Based on chat history	Based on vector search
Maintains context	Provides knowledge

Why can't we store unlimited chat history?

Unlimited history increases token usage, slows responses, and raises API costs.

How does Conversation Memory improve chatbot performance?

It enables the chatbot to understand follow-up questions, maintain context, and provide more natural conversations.

🎯 Tasks for Day 78

🧠 Theory

1. Define Conversation Memory.
2. Explain Buffer Memory and Window Memory.
3. Differentiate Memory and Retrieval.
4. Why is Chat History important?
5. List four real-world applications of Conversation Memory.

💻 Practical Task 1

Create a chatbot using **ConversationBufferMemory**.

💻 Practical Task 2

Ask five related questions and verify that the chatbot remembers previous messages.

💻 Practical Task 3

Replace **ConversationBufferMemory** with **ConversationBufferWindowMemory(k=5)** and compare the outputs.

💻 Practical Task 4

Create a conversation where the chatbot remembers the user's name, city, and profession.

💻 Practical Task 5

Compare the following memory types:

- Buffer Memory
- Window Memory
- Summary Memory
- Entity Memory

Include their advantages, disadvantages, and suitable use cases.